AWANTIKA SRIVASTAVA

AI / Machine Learning Engineer | Data Science | LLM & Computer Vision

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PROFILE SUMMARY

Machine Learning Engineer and Data Scientist with **2+ years of experience developing production-grade AI systems and machine learning pipelines**. Skilled in **Python, SQL, PyTorch, TensorFlow, and Scikit-learn** for building predictive models, NLP pipelines, and computer vision solutions. Experienced in **feature engineering, exploratory data analysis, anomaly detection, and model deployment** in real-world environments. Strong background in **LLM applications, RAG systems, OCR-based document processing, and real-time ML inference systems**.

CORE TECHNICAL SKILLS

- **Programming Languages & Tools:** Python, Numpy, Pandas, Matplotlib, Scikitlearn, C++, SQL
- **Statistics & Mathematics:** EDA, Statistical Modeling, Hypothesis Testing, A/B Testing, Probability, Linear Optimization, Trend Analysis, Experimental Design.
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, Hyperparameter Tuning, Feature Engineering.
- **Deep Learning:** CNN, RNN, LSTM, Transformers (BERT), Transfer Learning, Model Fine-tuning, RAG, LLMs, PyTorch, TensorFlow, Keras
- **Computer Vision**: Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Image & Video Processing Pipeline, Bounding Boxes & Confidence Scoring, Geometric Transformation, OpenCV, Paddle OCR
- **NLP & LLM:** RAG Systems, Prompt Engineering, Semantic Search, Text Processing, Transformer-based NLP, LLM Integration
- **MLOps & Deployment:** Docker, Flask APIs, Git, CI/CD, Model Monitoring
- **Cloud & Platforms:** AWS EC2, Linux, Jupyter Notebook

EXPERIENCE

Machine Learning Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Designed and deployed **machine learning models for production environments**, including training, evaluation, and performance monitoring.
- Developed **data pipelines and feature engineering workflows** to process large **structured and unstructured datasets** for machine learning tasks.
- Built **NLP pipelines for processing unstructured text data**, enabling automated extraction and analysis of textual information.
- Implemented **OCR-based document processing systems** to extract structured data from scanned documents and PDFs.
- Developed **document parsing pipelines** for automated information extraction and validation from enterprise documents.
- Designed **LLM-powered applications and RAG (Retrieval Augmented Generation) systems** for document search and question answering.
- Implemented **prompt engineering strategies and evaluation frameworks** to improve the accuracy and reliability of LLM responses.
- Built **machine learning APIs and microservices** to serve model predictions in production environments.
- Conducted **exploratory data analysis (EDA)** to identify patterns, anomalies, and insights from commercial datasets.
- Collaborated with **data engineers and product teams** to integrate ML pipelines and ensure scalable production deployment.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed** and deployed a real-time computer vision–based ML system to **detect** unsafe driver behaviors from **continuous video streams**.
- Trained and optimized **CNN-based object detection** model (SSD **MobileNet** architecture) to perform **real-time inference** on video data.
- Implemented **end-to-end ML pipelines** covering data ingestion, **preprocessing**, **model training**, **evaluation**, and **production** inference.
- Achieved **20–25 FPS** real-time processing with **<150** ms inference latency by optimizing modelsfor deployment.
- Built and integrated a **Flask-based web interface** to **visualize** detections and automatically **recorded 30-second** event clips, **reducing** manual review effort.

Amazon Stock Price prediction | Applied ML Project

- Designed real-time surveillance system to detect and track unauthorized objects/persons in restricted zones.
- Implemented **YOLO-based object detection pipeline using OpenCV** for continuous video stream analysis.
- Applied **bounding box filtering and confidence threshold tuning** to reduce false positives.
- Implemented geometric zone mapping and spatial reasoning to detect boundary violations.
- **Optimized** detection workflow using **Non-Max Suppression (NMS)** and **IoU-based** filtering.
- **Evaluated** model **performance** using **precision**, **recall**, and **mAP** metrics.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Deployed transformer-based **NLP models** as **production-ready** services with **REST APIs**.
- **Fine-tuned** and served a **DistilBERT**-based **sentiment** classification model for large-scale text inference.
- **Built** and deployed an interactive **streamlit web application** to perform real-time **sentiment analysis** on YouTube comments.
- **Processed** high-volume text **data** with sub-second inference **latency** for real-time sentiment analysis.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology